

Domain Density Dependence in 2FeFET-Based TCAM Bit-Cell: Search Failure Analysis and Design Optimization

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Abstract—This work investigates the effect of domain density (N_{Dom}) on FeFET variability, from the device level to the circuit level, using a 2FeFET-based ternary content addressable memory (TCAM) bit-cell. Monte Carlo simulations revealed that increasing N_{Dom} suppressed DTD variations in threshold voltage (C_v decreased by 80%) and memory window (C_v decreased by 79%). Furthermore, circuit-level analysis showed TCAM search failure probability exceeded 80% when $N_{Dom} < 100/\mu m^2$, while $N_{Dom} > 700/\mu m^2$ ensured robustness by stabilizing the dominant FeFET path. Finally, the proposed cross-scale design-space exploration framework formulates three quantitative guidelines that map the interplay between domain density, device sizing, and search functionality, providing a solid physical foundation for reliable 2FeFET-based TCAM design.

Index Terms—Domain density, ferroelectric field effect transistors (FeFETs), search failure, ternary content addressable memory (TCAM)

I. INTRODUCTION

NON volatile memory (NVM) technologies such as resistive RAM (RRAM), phase change memory (PCM), ferroelectric RAM (FeRAM), and magnetic tunneling junction (MTJ) are widely used in data storage, neuromorphic computing, and edge AI applications [1]–[4]. Ferroelectric field-effect transistors (FeFETs) are promising NVM candidates thanks to CMOS compatibility, low standby power, fast switching, and good scalability [5]. These characteristics make FeFETs suitable for a variety of energy-efficient applications including advanced sensor networks, embedded systems in autonomous vehicles, and real-time data processing for industrial automation [6], [7]. Despite CMOS compatibility, the wide adoption of FeFETs is hindered by reliability issues, particularly device-to-device (DTD) variations of up to 40% in the memory window (MW), which stem from the stochastic nature of domain switching in polycrystalline ferroelectric (FE) materials, including polar-axis orientation, grain inhomogeneity, and phase [8]–[14]. Beside, the number of FE domains (N_{Dom}) is also a significant factor affecting the DTD variations and stochastic behaviors. It can be tuned by engineering the grain size of the HfO_2 ferroelectric layer through process parameters like annealing, doping, and deposition, enabling controllable

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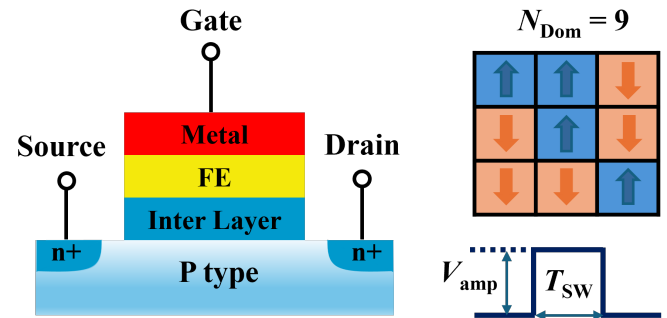


Fig. 1. Schematic of the Si-based FeFET structure and the N_{Dom} , which is the sole variation source considered in this study.

domain statistics and practical scalability. Fig. 1 shows the typical FeFET structure with the schematic of N_{Dom} and applied pulse waveform.

Compared with NAND-based TCAMs that rely on peripheral control and multistep sensing, the 2FeFET-based TCAM enables single cycle search with lower latency and simpler operation [15], [16]. Using two selector free FeFETs to encode ‘0’, ‘1’, and ‘X’, it achieves compact, nonvolatile storage with reduced power and better scalability than conventional NAND TCAMs. FeFET-based TCAM enables fast and energy-efficient associative search by exploiting the nonvolatile polarization states of ferroelectric materials. With compact cell designs and support for ‘0’, ‘1’, and ‘X’ states, it achieves one-cycle parallel search while reducing static power and area compared to SRAM-based TCAM, making it suitable for scalable, in-memory computing applications.

Nowadays, although research on FeFETs has become relatively mature, most studies focus separately on device level switching physics or circuit level compact modeling [17]–[20], and the quantitative linkage between N_{Dom} and DTD variations at the circuit level remains insufficiently explored. In this work, a physics-based FeFET model combined with Monte Carlo simulations [21], [22] is employed to analyze DTD variations induced by domain density. Based on this framework, we establish a 2FeFET-based TCAM bit-cell and investigate how these variations influence circuit level behavior, providing physics-guided design insights for optimizing domain density toward more reliable FeFET-based TCAMs.

The rest of this paper is organized as follows. Section II presents our proposed comprehensive device-circuit co-simulation framework. Section III-A discusses our model validation results. Section III-B describes the TCAM bit-cell implementation and analyzes potential failure mechanisms. Next, the benchmarking of our model is presented in Sec-

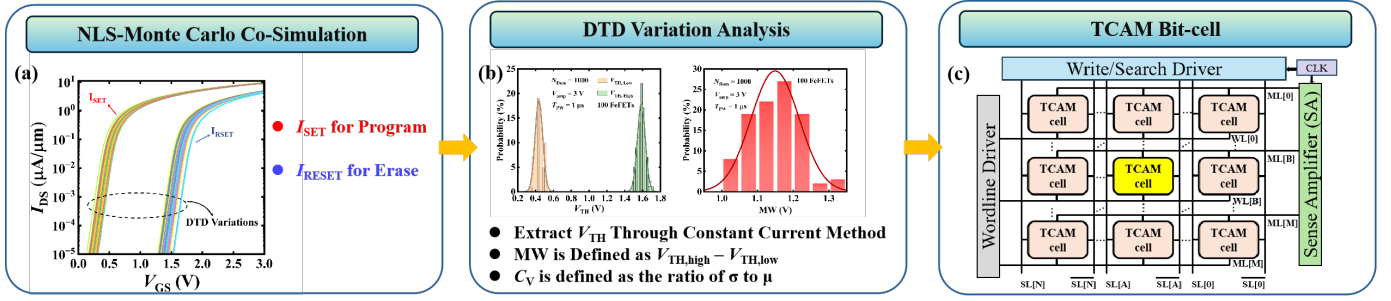


Fig. 2. (a) Based on the NLS FeFETs model combined with Monte Carlo simulations, the transfer characteristic curves for both set and reset operations were obtained, from which the DTD variations were extracted. (b) Flowchart illustrating the modeling framework used in this study. (c) Schematic of a TCAM array composed of 2FeFET-based cells, used to evaluate domain-density-induced search failures at the circuit level.

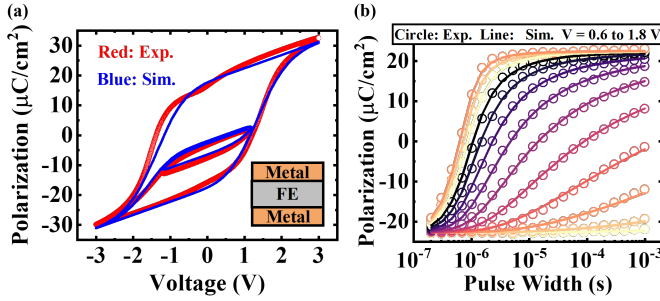


Fig. 3. (a) Experimental and simulated hysteresis loops of polarization with respect to voltage for a single FE capacitor (b) Experimental and simulated polarization-pulse width curves under 0.6V to 1.8V, showing ferroelectric switching dynamics [21].

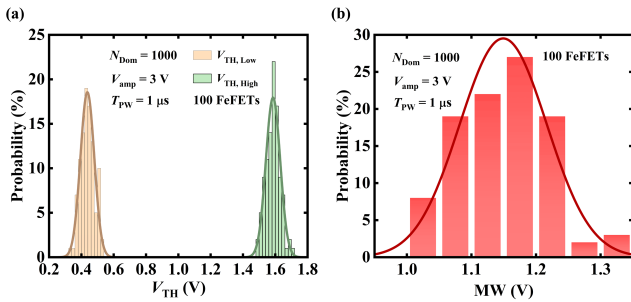


Fig. 4. (a) Simulated threshold voltage distributions of $V_{TH,high}$ and $V_{TH,low}$ across 100 Monte Carlo FeFET instances, along with corresponding Gaussian fits. (b) Simulated distribution of MW with Gaussian fit, based on 100 Monte Carlo FeFET simulations.

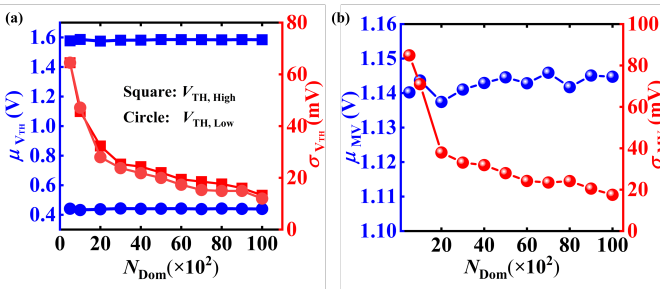


Fig. 5. Dependence of V_{TH} statistics and MW metrics on N_{Dom} . (a) Mean ($\mu_{V_{TH}}$, left axis) and standard deviation ($\sigma_{V_{TH}}$, right axis) for V_{TH} states. (b) Mean (μ_{MW} , left axis) and standard deviation (σ_{MW} , right axis) of the MW.

tion III-C before Section IV concludes this paper.

II. DEVICE-CIRCUIT CO-SIMULATION FRAMEWORK

This study first employs a Verilog-A implemented FeFET model, calibrated against experimental polarization data, and combines it with Monte Carlo simulations to systematically study the impact of N_{Dom} on DTD variations. The calibrated circuit-compatible model is then applied in Cadence Virtuoso to simulate the proposed 2FeFET-based TCAM bit-cell and analyze the influence of domain density on search operation. The whole device-circuit co-simulation framework is illustrated in Fig. 2. Furthermore, to validate the effectiveness of the model and extend it to practical circuit design, we calibrated both the dynamic response and the hysteresis loop for a single FE capacitor using experimental measurements shown in Fig. 3(a), which corresponds to a stand-alone FE capacitor rather than a FeFET, and Fig. 3(b) [21].

Existing Monte Carlo and kinetic Monte Carlo FeFET studies mainly focus on device-level switching statistics and DTD variation from finite domains [21], [22], while SPICE-compatible models reproduce terminal I - V , hysteresis, charge matching, and circuit response under practical operating conditions and design constraints [17], [18], [23]. Previous FeFET TCAM studies have primarily addressed cell topology, energy efficiency, and experimental feasibility [24], [25].

In contrast, this work treats N_{Dom} as a process-related statistical variable and propagates the resulting FeFET V_{TH} /MW variation to match-line behavior, search-failure probability, and domain-density design guidelines in a 2FeFET TCAM bit-cell. Thus, the main contribution is a cross-scale reliability mapping from domain-density-induced FeFET variability to TCAM-level search failure and design guidelines.

III. RESULTS AND DISCUSSION

The following analysis links the device-level N_{Dom} -dependent V_{TH} /MW statistics to TCAM match-line latency, failure boundary, and domain-density design rules.

A. Device Variability Analysis

Fig. 2(a) showed the simulated I - V curves with set and reset device operations of FeFETs and reflects the obvious DTD variations through Monte Carlo simulations. For each

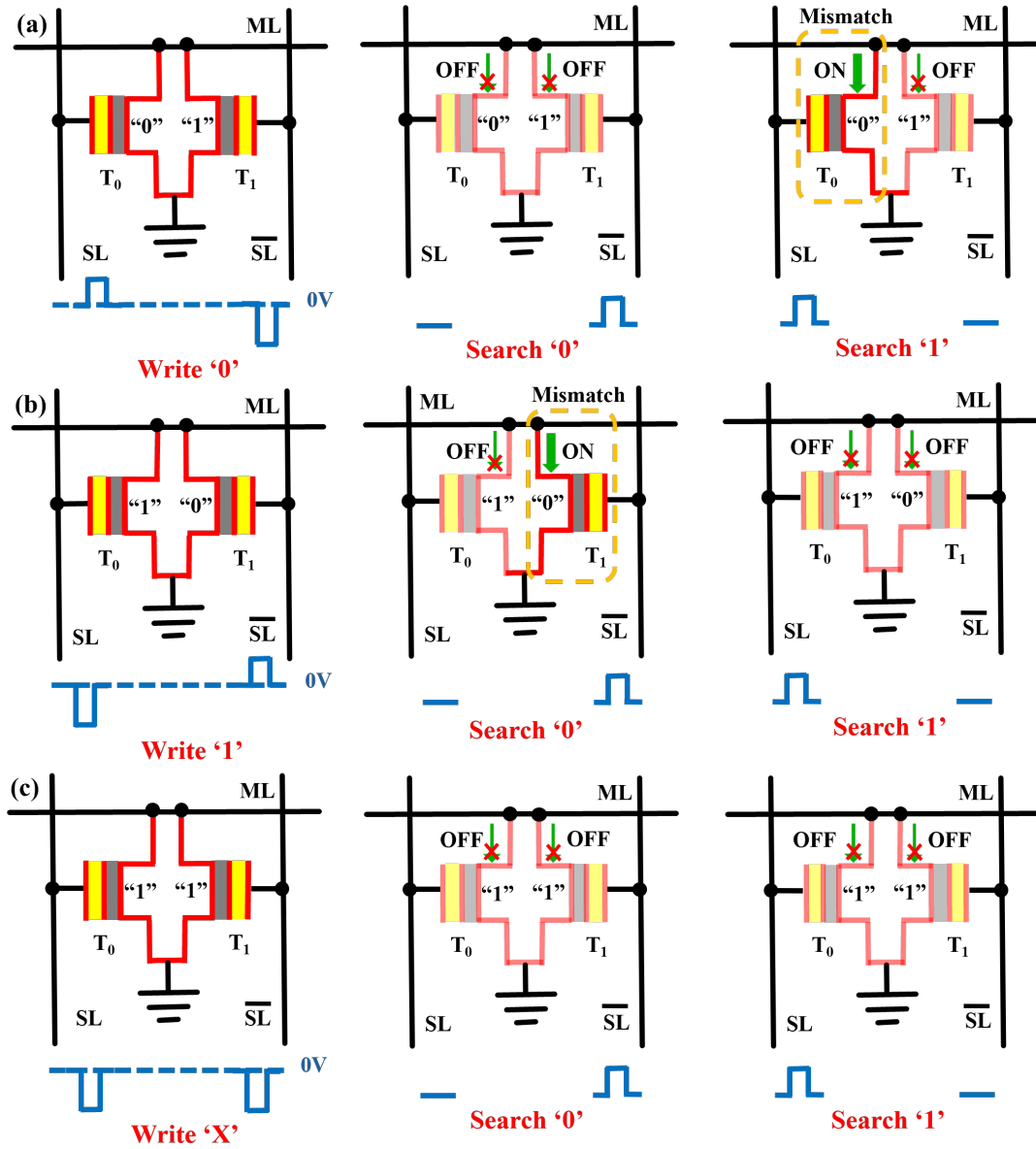


Fig. 6. Waveform and switching behavior of the 2FeFET-based TCAM bit-cell during write and search operations, where (a) illustrates the case of writing '0' followed by searches for '0' and '1', (b) shows writing '1' followed by searches for '0' and '1', and (c) depicts writing 'X' followed by searches for '0' and '1'.

parameter configuration, simulations were performed on 100 independent FeFET devices in Fig. 2(a), with the gate voltage swept from -0.5 V to 3 V and a fixed drain bias of 0.05 V. We extracted $V_{TH,high}$ and $V_{TH,low}$ at $I_d = 10nA$, and the MW was defined as their difference.

C_v , which is defined as the ratio of the standard deviation (σ) to the mean (μ) of the respective parameter, was adopted as a quantitative metric to characterize DTD variations.

To quantify the impact of N_{Dom} on DTD variations, we simulated FeFETs across a wide range of N_{Dom} values. Fig. 4(a) and Fig. 4(b) presented the distributions of $V_{TH,high}$, $V_{TH,low}$, and MW, showing their statistical evolution with increasing domain density. Joint histogram analysis revealed how threshold separation scales with domain configuration, and histogram fitting yields the C_v , linking device structure

and operating conditions to DTD trends [26], [27]. As demonstrated in Fig. 5(a) and Fig. 5(b), N_{Dom} plays a key role in DTD control through statistical averaging of ferroelectric switching events. Increasing N_{Dom} produces a clear monotonic decrease in C_v for both $V_{TH,high}$ and $V_{TH,low}$, with the latter slightly higher due to its proximity to the switching threshold and greater sensitivity to incomplete domain transitions. The corresponding drop in C_v of MW confirms the stabilizing effect of larger N_{Dom} . In the 2FeFET-based TCAM bit-cell, a lower C_v of MW ensures tighter threshold separation and higher search reliability. This underscores the importance of domain engineering not only in N_{Dom} , but also in distribution uniformity and scaling integration for next-generation development.

In summary, increasing N_{Dom} suppresses FeFET V_{TH}/MW

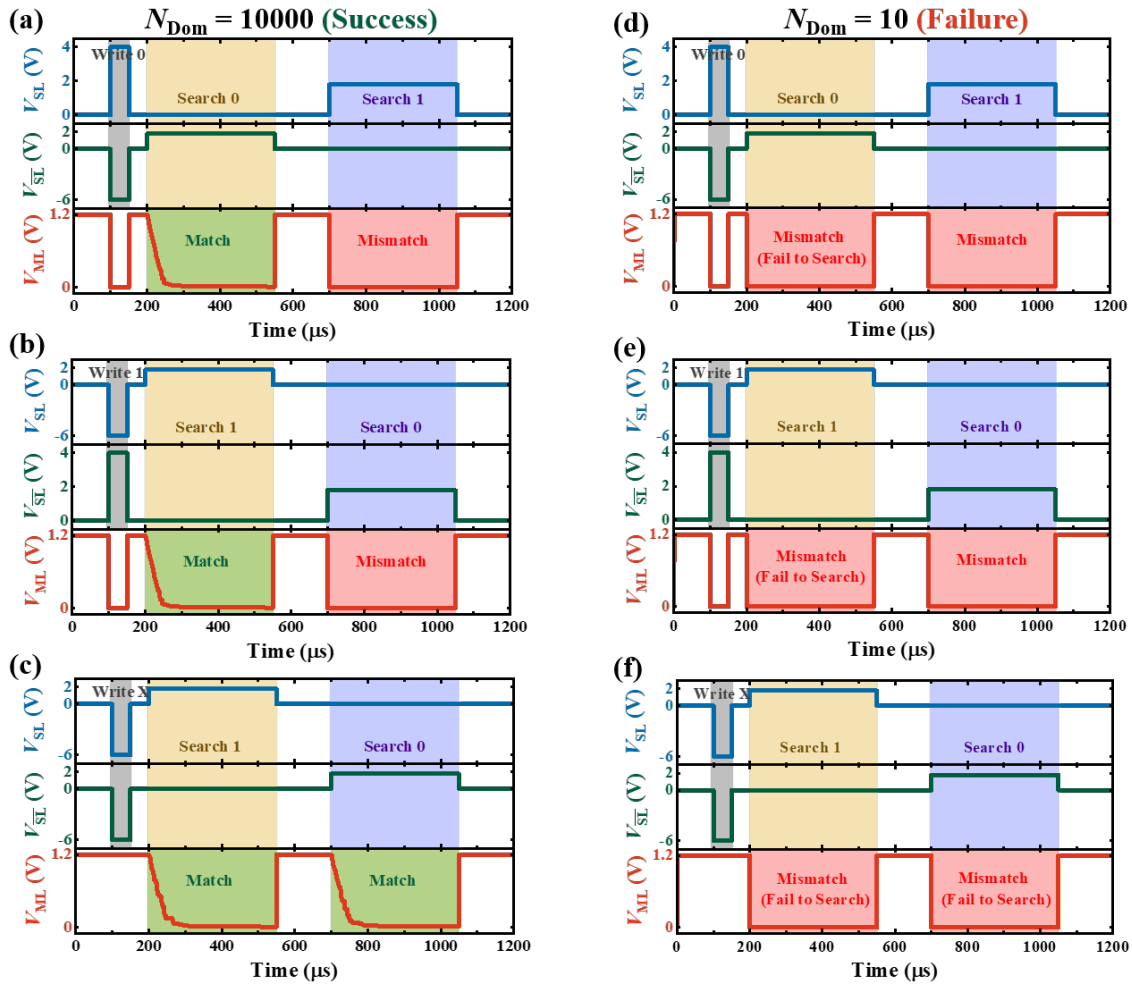


Fig. 7. Search-line and match-line waveforms of the 2FeFET-based TCAM bit-cell under different N_{Dom} , where (a–c) show successful matches for $N_{Dom}=10000$ during write ‘0’, write ‘1’, and write ‘X’ operations followed by searches for ‘0’ and ‘1’, and (d–f) show the corresponding failed matches for $N_{Dom}=10$.

variability, which improves the separation between conductive and nonconductive branches in the TCAM bit-cell and reduces the risk of premature match-line discharge.

B. TCAM Implementation and Failure Analysis

We constructed and illustrated a 2FeFET-based TCAM bit-cell circuit to investigate the impact of FeFET properties on matchline behavior and failure dynamics [24], [25], [28], [29]. In the TCAM search operation, the matchline (ML) is precharged as the sensing node, while SL and $\overline{SL}$ serve as complementary search lines that apply the query data to the two FeFET branches. During operation, gate voltages were applied to T0 and T1 to polarize the FeFETs into logic ‘0’, ‘1’, or a fuzzy ‘X’ state. During the search operation, differential voltages were applied to SL and $\overline{SL}$, and the matchline discharge behavior reflected the correctness of the stored–search comparison. Matched cases produce slower ML discharges because the effective discharge path is suppressed, whereas mismatches induce faster voltage drops due to the formation of a conductive discharge path. The cell schematic

TABLE I
PULSE SUMMARY FOR WRITING AND SEARCHING OF 2FEFET BASED TCAM BIT-CELL

Operation	T_0	T_1	SL	$\overline{SL}$
Write ‘0’	‘0’	‘1’	4 V, 50 μ s	-6 V, 50 μ s
Write ‘1’	‘1’	‘0’	-6 V, 50 μ s	4 V, 50 μ s
Write ‘X’	‘1’	‘1’	-6 V, 50 μ s	-6 V, 50 μ s
Search ‘0’	-	-	0 V, 0.35 ms	1.8 V, 0.35 ms
Search ‘1’	-	-	1.8 V, 0.35 ms	0 V, 0.35 ms

was shown in Fig. 6, and the write/search bias conditions were summarized in Table. I.

To evaluate the match-line response, the collapse time is extracted using the 90%–10% transition interval, defined as $t_{90-10} = t(V_{ML} = 0.1V_{ML,0}) - t(V_{ML} = 0.9V_{ML,0})$. This interval follows the standard fall-time or transition-duration definition for falling waveforms and is less sensitive to local waveform perturbations than a single-threshold crossing [30], [31]. In the TCAM sensing context, the 90% point avoids treating minor precharge settling or search-line coupling as a true discharge event, while the 10% point corresponds to an unambiguous low match-line level. Representative successful

and failed waveforms obtained using this method are shown in Fig. 7.

A case with $t_{90-10} < 10 \mu\text{s}$ is classified as an early-collapse event. This threshold is not intended to define a universal TCAM access time. Instead, it serves as a conservative guard-band within the 0.35 ms search pulse used in this work, where $10 \mu\text{s}/0.35 \text{ ms} = 2.86\%$. The physical meaning of this criterion can be understood from a first-order match-line discharge approximation,

$$V_{\text{ML}}(t) = V_{\text{ML},0} \exp\left(-\frac{t}{\tau_{\text{ML}}}\right), \quad t_{90-10} = \ln(9)\tau_{\text{ML}}. \quad (1)$$

Here, τ_{ML} is the effective discharge time constant of the match line under the evaluated search condition. A smaller τ_{ML} indicates a stronger discharge path. Therefore, $t_{90-10} < 10 \mu\text{s}$ corresponds to $\tau_{\text{ML}} < 10 \mu\text{s}/\ln(9) \approx 4.55 \mu\text{s}$, indicating that the nominally high-resistance match path has degraded into a strong discharge path. Under this condition, the match line cannot maintain an adequate sensing margin for reliable match detection. Using this criterion, we swept a broad range of FE N_{Dom} to assess how domain allocation shapes discharge dynamics and failure susceptibility.

C. Design Space Exploration

To investigate how domain allocation impacted matchline behavior in a 2FeFET-based TCAM bit-cell, we performed a latency-driven exploration of two representative write schemes (Table. I). Each bit-cell comprised two FeFETs, T_0 and T_1 , which were programmed to complementary threshold states: storing '0' set T_0 to '0' and T_1 to '1', whereas storing '1' applied the opposite.

Fig. 8 (a) and Fig. 8 (b) presented heatmaps of matchline latency as a function of the N_{Dom} assigned to T_0 and T_1 . The purple regions (latency $< 10 \mu\text{s}$) were classified as quasi-failure regions, where potential reliability issues may arise during measurement due to process variations or operational uncertainties, potentially leading to functional failure. In contrast, the regions with latency $> 10 \mu\text{s}$ were labeled as *success* regions, representing parameter configurations that should be prioritized in practical circuit design to ensure stable and reliable matchline discharge behavior.

Representative time domain waveforms for both successful and failed comparisons are shown in Fig. 7. The left column illustrates correct operation when the stored and search data matched, whereas the right column shows early-collapse cases characterized by an immediate match-line voltage drop. These zero-latency traces represent the limiting cases of the early-collapse behavior captured by the $t_{90-10} < 10 \mu\text{s}$ criterion, rather than a separate failure threshold.

The matchline latency behavior differs notably between writing '0' and '1'. For the '0' case (Fig. 8 (a)), latency is primarily influenced by T_1 , with acceptable performance observed when T_1 has more than $700/\mu\text{m}^2$, regardless of T_0 allocation. This indicates that T_1 polarization dominates discharge dynamics, and increasing T_1 's domain count can offset a reduced T_0 . However, when both T_0 and T_1 have fewer than $100/\mu\text{m}^2$, the discharge becomes too rapid, leading

to early voltage collapse. In contrast, for the '1' case (Fig. 8 (b)), latency strongly depends on T_0 , and assigning more than $300/\mu\text{m}^2$ to T_0 significantly increases latency. Here, adequate T_0 polarization prevents premature discharge, rendering the impact of T_1 relatively minor once T_0 is sufficiently provisioned.

Monte Carlo simulations elucidated the timing failure mechanism in greater detail. When the FE domain density is insufficient, the intrinsic hysteresis loop of the FeFET collapses, eliminating V_{TH} separation between the high and low polarization states. In this degraded regime the matchline discharges almost instantaneously, and TCAM searches inevitably fail. In summary, the search failure originates from the loss of threshold-state separation under insufficient domain density, which turns the nominally high-resistance branch into a conductive discharge path.

Fig. 9 provided a detailed analysis of the quasi-failure regime. In this figure, red denoted actual failure cases where the matchline latency was zero, whereas green denoted near-failure events characterized by small but nonzero latency. A high degree of randomness was observed in this region when N_{Dom} was small, reinforcing the notion that TCAM search operations became increasingly complex and unpredictable under insufficient domain provisioning. Monte Carlo results (Fig. 10) quantified 81.25% search failure when $N_{\text{Dom}} < 100/\mu\text{m}^2$, dropping below 5% when $N_{\text{Dom}} > 500/\mu\text{m}^2$. This transition correlates with σ_{MW} decreasing below approximately 90 mV (Fig. 5 (b)). These results highlight the necessity of designing TCAM bit-cell within Region I of Fig. 8, where a sufficiently large N_{Dom} ensures robust and deterministic matchline behavior.

Results show that asymmetric domain allocation impacts matchline latency. The dominant FeFET defines the stored logic, and search failures (latency $< 10 \mu\text{s}$) mainly occur when both devices have low N_{Dom} . Increasing the dominant device's N_{Dom} yields a transition near 300–500, indicating that allocating device-level domain redundancy by increasing the effective number of statistically independent switching domains to the discharge-controlling device reduces latency variations and enhances TCAM reliability.

The simulated N_{Dom} is treated as an effective density of statistically independent switching domains rather than a strict crystallographic grain density. To relate it to manufacturability, the effective switching-domain pitch is estimated as

$$d_{\text{eff}} \approx \frac{1000}{\sqrt{N_{\text{Dom}}}} \text{ nm}, \quad (2)$$

where N_{Dom} is expressed in μm^{-2} . Thus, $N_{\text{Dom}} = 700 \mu\text{m}^{-2}$ corresponds to $d_{\text{eff}} \approx 37.8 \text{ nm}$. This value should be viewed as an effective switching-domain pitch rather than a strict grain-size limit, since statistically independent switching units in polycrystalline HfO_2 -based films do not necessarily coincide with individual grains. Reported HfO_2 -based FeFET studies have linked domain statistics, switching uniformity, crystalline-phase nonuniformity, scaled-device process variations, and DTD variability [11], [13], [22]. Therefore, low- N_{Dom} cases represent pessimistic process cor-

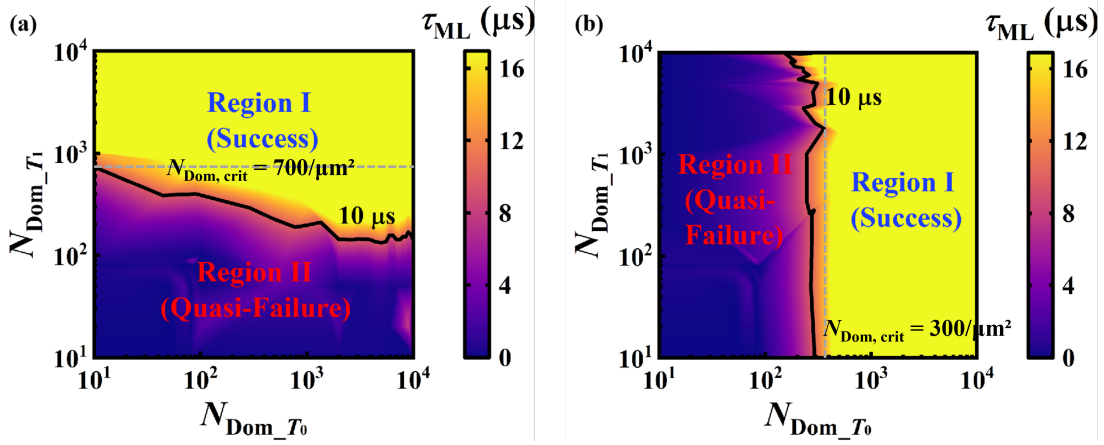


Fig. 8. (a) Latency landscape for the *Search 0* operation. Yellow denotes latency $> 10 \mu s$, whereas purple indicates latency $< 10 \mu s$. The black contour traces the $10 \mu s$ boundary. (b) Latency landscape for the *Search 1* operation. The color scale and $10 \mu s$ contour follow the convention in Fig. 8(a). Yellow and purple regions are separated by the $10 \mu s$ latency threshold (black contour). Critical N_{Dom} for T1 (*Search 0*) and T0 (*Search 1*) are marked at $700/\mu m^2$ and $300/\mu m^2$, respectively.

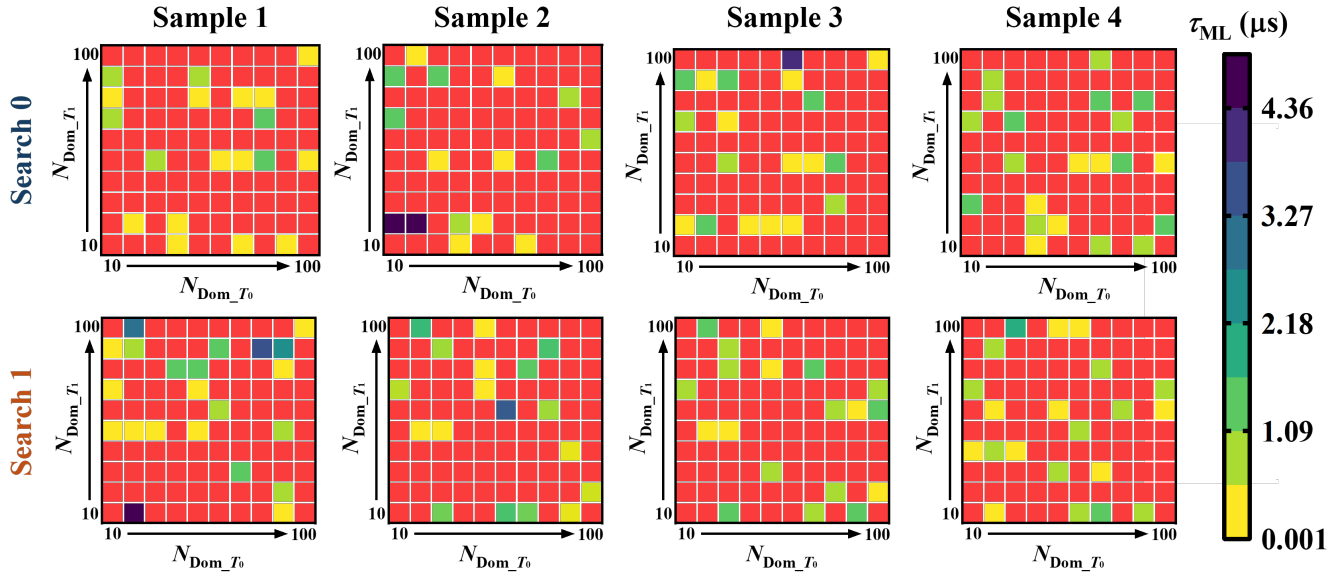


Fig. 9. Matchline latency distribution for search 0 and 1, including search failure (red) and search success (other colors).

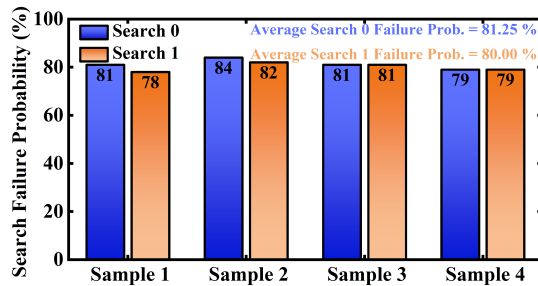


Fig. 10. Search failure probability for logic '0' and '1' across different samples. The average failure rates are 81.25% for search 0 and 80.00% for search 1.

ners, while $N_{Dom} \approx 700 \mu m^{-2}$ provides a conservative manufacturability-oriented target for reliable TCAM operation.

Based on the device- and circuit-level findings presented

above, we distill three practical guidelines to minimize DTD variations and improve TCAM search reliability:

- Minimum $N_{Dom} = 700/\mu m^2$ for FeFETs on discharge-critical path (T1 for stored "0", T0 for "1")
- Asymmetric domain allocation: Prioritize N_{Dom} in dominant FeFET (Fig.8)
- Process control target: Maintain an effective switching-domain density of approximately $N_{Dom} \approx 700 \mu m^{-2}$, corresponding to an effective domain pitch of about 38 nm, to suppress early match-line collapse.

IV. CONCLUSION

In this work, the impact of N_{Dom} on 2FeFET-based TCAM bit-cells is analyzed through Monte Carlo simulations. At the device level, increasing N_{Dom} reduces the variability of both the memory window and threshold voltage. At the circuit level, higher N_{Dom} lowers the failure probability, and

TCAM functionality remains stable when the dominant device has a ferroelectric domain density above $700/\mu\text{m}^2$. These results highlight the effectiveness of domain engineering for enhancing search accuracy at the bit-cell level, indicating that optimized domain allocation is essential for achieving reliable and energy-efficient FeFET-based TCAM cells.

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